INCH-POUND

MS9381A
W/ AMENDMENT 2
10 June 2025
SUPERSEDING
MS9381A
W/ AMENDMENT 1
5 November 2024

DETAIL SPECIFICATION

COUPLING, TUBE - BRAZING, SAE AMS5646

Inactive for new design after 14 August 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.

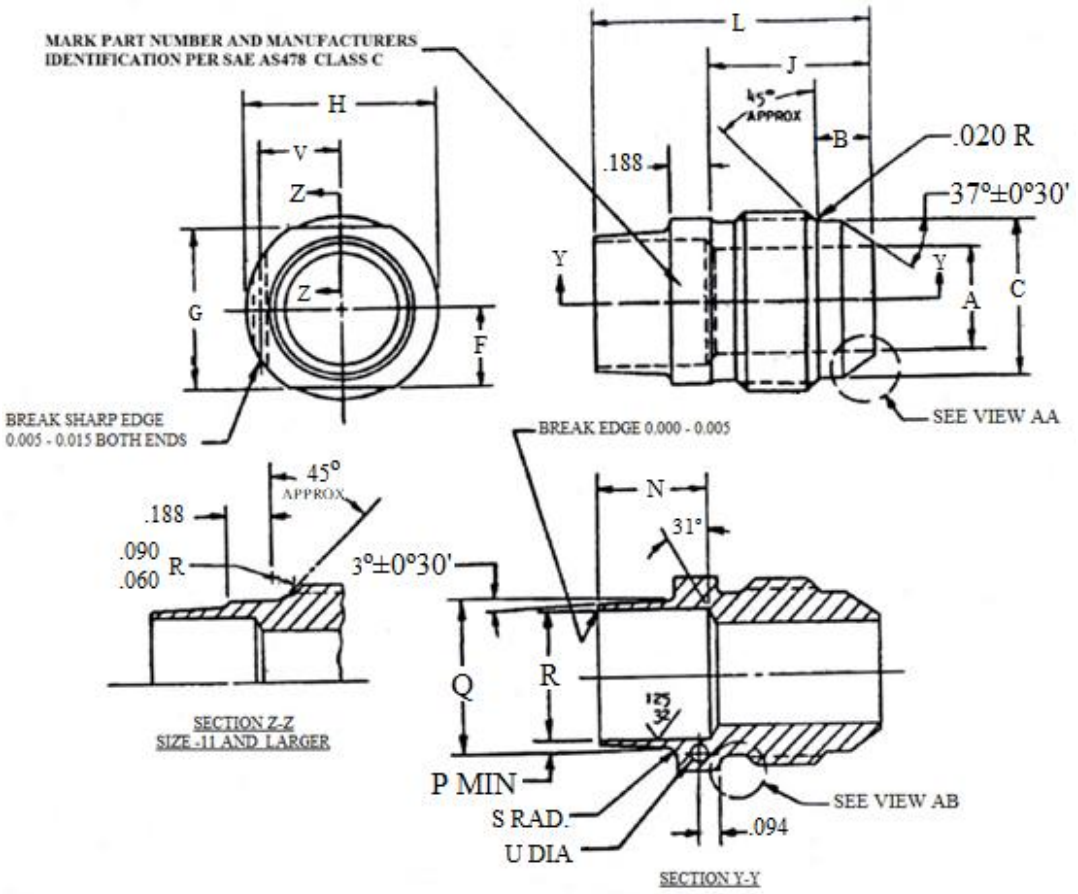


FIGURE 1. Item drawing.

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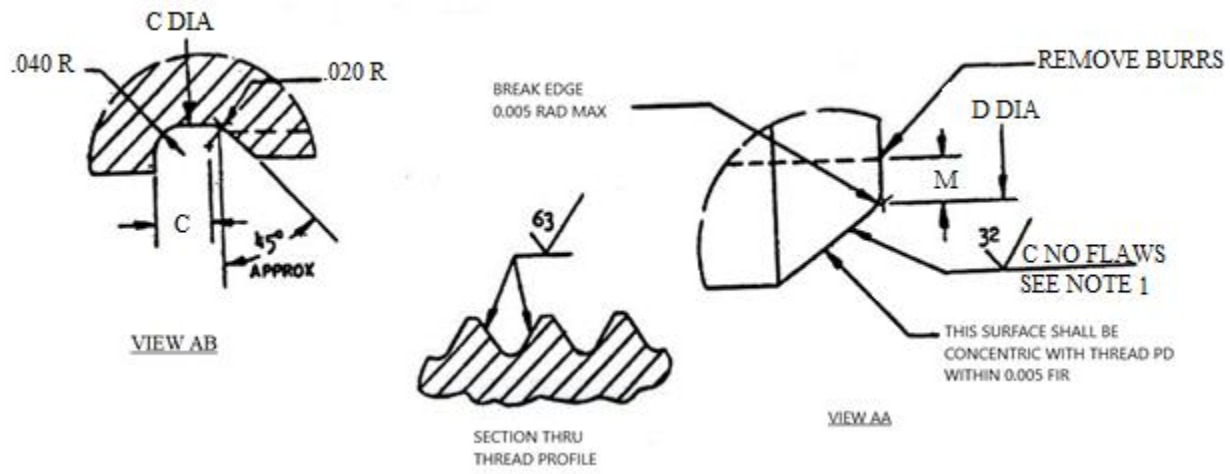


FIGURE 1. Item drawing – Continued.

Requirements:

1. Conical surface shall be free of nicks, chatter marks or low spots when viewed with the unaided eye.
2. All diameters shall be concentric within .010 FIR unless otherwise specified.
3. Dimensions are to sharp corners, unless otherwise specified.
4. Material: corrosion resistant steel, SAE AMS5646.
5. Parts shall be degreased and then passivated in accordance with SAE AMS2700, Method 1, Type 6.
6. Surface roughness: SAE AS291. Unless otherwise specified, surfaces to be 125 microinches.
7. Break sharp edges .003 - .015, unless otherwise specified.
8. Dimensions in inches. Unless otherwise specified, tolerances: linear dimensions ±0.010, angular dimensions ±5º.
9. Do not use unassigned part numbers.

Reference Documents. This document references the following:

SAE AMS2700	SAE AS291
SAE AMS5646	SAE AS478

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notatations.

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TABLE I. Part number dimensions.

Part No.	Nom Tube OD	Thread T					A	B +0.015 -0.010	C Dia +.000 -0.010	D Dia ±.003	E +0.015 -.000	F	G ±.005
		Size	Major Dia	Pitch Dia	Minor Dia	Root Rad							
MS9381-02	0.125	.3125-24UNJF-3A	.3053-.3125	.2827-.2854	.2591-.2644	.0063-.0075	.058-.065	0.144	0.245	0.132	0.063	0.125	0.25
MS9381-03	0.188	375-24UNJF-3A	3678-3750	3450-3479	3214-3268	.0063-.0075	121-.128	0.144	307	0.195	0.063	0.156	312
MS9381-04	0.25	4375-20UNJF-3A	.4294-.4375	.4019-.4050	.3736-.3797	.0075-.0090	168-.175	160	359	242	0.075	0.188	375
MS9381-05	0.312	500-20UNJF-3A	.4919-.5000	4643-.4765	.4360-.4422	.0075-.0090	230-.237	0.16	0.421	0.304	0.075	0.219	0.438
MS9381-06	0.375	5625-18UNJF-3A	5538-.5625	5230-.5264	.4916-.4983	.0083-.0100	293-.301	0.165	0.476	0.368	0.083	250	0.5
MS9381-07	0.438	625-18UNJF-3A	.6163-.6250	5854-.5889	5540-.5608	.0083-.0100	355-363	165	0.539	0.43	0.083	281	0.562
MS9381-08	0.5	750-16UNJF-3A	7406-7500	.7056-.7094	.6702-.6778	.0094-.0113	.387-.395	0.229	0.654	0.462	0.094	0.344	0.688
MS9381-09	0.562	8125-16UNJ-3A	.8031-.8125	.7683-.7719	.7329-.7403	.0094-.0113	.434-.442	0.229	0.716	0.509	0.094	0.375	0.75
MS9381-10	0.625	875-14UNJF-3A	.8647-.8750	.8245-.8286	.7841-.7925	.0107-.0129	.480-.488	0.255	0.767	0.555	0.107	0.391	0.781
MS9381-11	0.688	1.000-12UNJF-3A	.9886-1.0000	.9415-.9459	.8944-.9038	.0125-.0150	.542-.552	0.304	0.875	0.619	0.125	0.438	0.875
MS9381-12	0.75	1.0625-12UNJ-3A	1.0511-1.0625	1.0042-1.0084	.9570-.9663	.0125-.0150	.604-.614	0.304	0.938	0.681	0.125	0.469	0.938
MS9381-14	0.875	1.1875-12UNJ-3A	1.1761-1.1875	1.1291-1.1334	1.0820-1.0913	.0125-.0150	.729-.739	0.304	1.062	0.806	0.125	0.531	1.062
MS9381-16	1	1.3125-12UNJ-3A	1.3011-1.3125	1.2541-1.2584	1.2070-1.2163	.0125-.0150	.839-.851	0.312	1.188	0.918	0.125	0.594	1.188
MS9381-18	1.125	1.500-12UNJF-3A	1.4886-1.5000	1.4411-1.4459	1.3940-1.4038	.0125-.0150	.948-.960	0.312	1.375	1.027	0.125	0.656	1.312
MS9381-20	1.25	1.625-12UNJ-3A	1.6136-1.6250	1.5665-1.5709	1.5193-1.5288	.0125-.0150	1.073-1.086	363	1.501	1.153	0.125	0.719	1.438
MS9381-24	1.5	1.875-12UNJ-3A	1.8636-1.8750	1.8164-1.8209	1.7692-1.7788	.0125-.0150	1.307-1.320	374	1.75	1.387	0.125	0.844	1.688
MS9381-28	1.75	2.250-12UNJ-3A	2.2386-2.2500	2.1914-2.1959	2.1442-2.1538	.0125-.0150	1.542-1.557	447	2.125	1.652	0.125	1	2
MS9381-32	2	2.500-12UNJ-3A	2.4886-2.5000	2.4413-2.4459	2.3941-2.4038	.0125-.0150	1.776-1.791	457	2.375	1.886	0.125	1.125	2.25
MS9381-40	2.5	3.000-12UNJ-3A	2.9886-3.0000	2.9412-2.9459	2.8940-2.9038	.0125-.0150	2.276-2.291	391	2.882	2.386	0.125	1.375	2.75
MS9381-48	3	3.500-12UNJ-3A	3.4886-3.5000	3.4411-3.4459	3.3940-3.4038	.0125-.0150	2.776-2.791	391	3.382	2.886	125	1.625	3.25

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TABLE I. Part Number Dimensions - Continued

Part No.	H Dia +.010 -.000	J	L	M Min	N	P Min	Q Dia +.010 -.000	R Dia +.002 -.000	S	U	V	Approx Weight Lb/100
MS9381-02	0.344	0.415	0.841	0.03	0.346	0.022	0.17	0.115	.031-.062	.045-.050	102-.112	1.17
MS9381-03	0.438	0.446	0.884	0.03	0.358	0.023	0.234	0.178	.031-.062	.057-.062	.139-.149	1.5
MS9381-04	0.5	0.517	0.973	0.03	0.376	0.026	0.303	0.24	.047-078	.057-.062	.170-.180	2.05
MS9381-05	0.593	0.517	0.973	0.03	0.376	0.029	0.37	0.302	.047-078	.065-.075	207-217	2.64
MS9381-06	0.656	0.523	0.979	0.03	0.376	0.029	0.433	0.365	.047-078	.065-.075	238-258	3.12
MS9381-07	0.719	523	1.006	0.03	0.403	0.033	0.504	0.428	.047-078	.065-.075	.270-.290	3.98
MS9381-08	0.781	0.633	1.145	0.03	0.432	0.034	0.568	0.489	.047-078	.065-.075	300-320	5.82
MS9381-09	0.844	0.633	1.164	0.03	0.451	0.038	0.637	551	.047-078	.065-.075	331-351	6.66
MS9381-10	904	0.747	1.306	0.03	0.479	0.039	0.703	0.614	.047-078	.065-.075	362-382	8.41
MS9381-11	1	0.853	1.443	0.03	0.51	0.041	0.769	0.677	.047-078	.065-.075	.410-.430	11.8
MS9381-12	1.094	0.853	1.443	0.03	0.51	0.041	0.831	0.739	.047-078	.065-.075	.440-.460	13.13
MS9381-14	1.188	0.898	1.488	0.03	0.51	0.041	0.956	0.864	.047-078	.065-.075	.503-.523	15.96
MS9381-16	1.312	908	1.498	0.03	0.51	0.046	1.091	0.989	.078-.109	.065-.075	565-.585	18.27
MS9381-18	1.5	0.954	1.544	0.03	0.51	0.046	1.216	1.114	.078-.109	.065-.075	.644-.664	22.33
MS9381-20	1.625	954	1.544	0.03	0.51	0.051	1.351	1.239	.078-.109	.065-.075	.706-.726	27.38
MS9381-24	1.875	1.079	1.669	0.03	0.51	0.061	1.616	1.484	.078-.109	.065-.075	.830-.850	38.13
MS9381-28	2.25	1.204	1.794	0.044	0.51	0.071	1.886	1.734	.078-.109	.065-.075	.986-1.006	61.64
MS9381-32	2.5	1.329	1.919	0.044	0.51	0.086	2.166	1.984	.078-.109	.065-.075	1.111-1.131	79.24
MS9381-40	3	1.105	1.695	0.044	0.51	0.086	2.666	2.484	.094-.125	1065-.075	1.361-1.381	82.11
MS9381-48	3.5	1.183	1.773	0.044	0.51	0.091	3.176	2.984	.094-.125	.065-.075	1.611-1.631	105.85

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CONCLUDING MATERIAL

Custodians:
Navy – AS
Air Force – 11

Preparing Activity:
Navy – AS

Review Activities:
DLA – CC

(Project: 4730 - 2025 - 040)

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